

Lecture 6

Introduction to JavaScript

Outlin

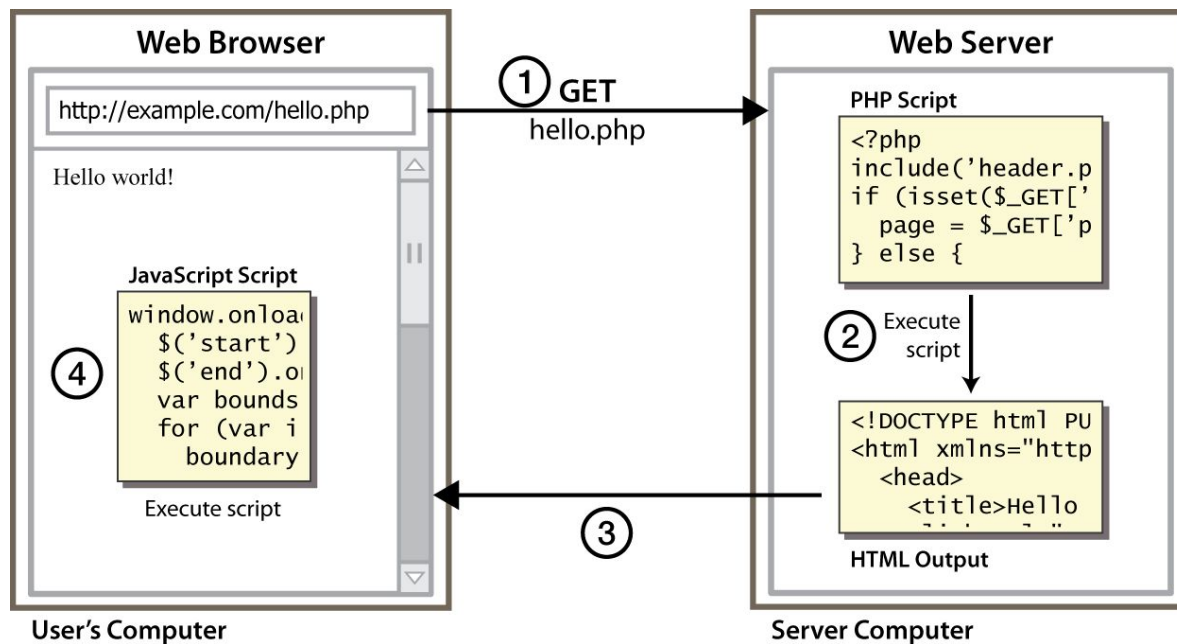
- **What is JavaScript?**
- **Embedding JavaScript with HTML**
- **JavaScript conventions**
- **Variables in JavaScript**
- **JavaScript operators**
- **Input output in JavaScript**
- **JavaScript functions**
- **Conditional Statements**
- **Looping Statements**

1.

- JavaScript is a **client-side** scripting language
- JavaScript was designed to add **interactivity** to HTML pages
- JavaScript is used in **millions** of Web pages to improve the design, validate forms, detect browsers, create cookies, and much more

Client-side scripting

client-side script: code runs in browser *after* page is sent back from server often this code manipulates the page or responds to user actions



Why use client-side programming?

PHP already allows us to create dynamic web pages. Why also use client-side scripting?

client-side scripting (JavaScript) benefits:

- usability:** can modify a page without having to post back to the server (faster UI)

- efficiency:** can make small, quick changes to page without waiting for server

- event-driven:** can respond to user actions like clicks and key presses

server-side programming (PHP) benefits:

- security:** has access to server's private data; client can't see source code

- compatibility:** not subject to browser compatibility issues

- power:** can write files, open connections to servers, connect to databases, ...

What is JavaScript?

- a lightweight programming language ("scripting language")
- used to make web pages interactive
 - insert dynamic text into HTML (ex: user name)
 - react to events (ex: page load user click)
 - get information about a user's computer (ex: browser type)
 - perform calculations on user's computer (ex: form validation)
- a web standard (but not supported identically by all browsers)
- NOT related to Java other than by name and some syntactic similarities

JavaScript vs. Java

- **interpreted**, not compiled



+



= JavaScript

- more relaxed syntax and rules
 - fewer and "looser" data types
 - variables don't need to be declared
 - errors often silent (few exceptions)
- key construct is the **function** rather than the class
 - "first-class" functions are used in many situations
- contained within a web page and integrates with its HTML/CSS content

JavaScript vs. PHP

- similarities:

- both are **interpreted**, not compiled
- both are relaxed about syntax, rules, and types
- both are case-sensitive
- both have built-in regular expressions for powerful text processing

- differences:

- JS is more object-oriented: `noun.verb()`, less procedural: `verb(noun)`
- JS focuses on UIs and interacting with a document; PHP on HTML output and files/forms
- JS code runs on the client's browser; PHP code runs on the web server

1.

- JavaScript is an **interpreted language** (means that scripts execute without preliminary compilation)
- JavaScript is usually **embedded** directly into HTML pages
- Everyone can use JavaScript without purchasing a license

1.

- **JavaScript is the **most popular** scripting language on the internet, and works in all major browsers, such as Internet Explorer, Mozilla, Firefox, Netscape, Opera**

1.1 JavaScript: Common

- JavaScript gives HTML designers a **programming tool**
- JavaScript can **react to events**
- JavaScript can **read and write** HTML elements
- JavaScript can be used to **validate data**
- JavaScript can be used to detect the **visitor's browser**
- JavaScript can be used to create **cookies**

2. Embedding JavaScript in

- There are **two** methods to **embed** JavaScript in to HTML code
 - **Internal Script:** directly written in HTML code
 - **External Script:** written in separate file
- **<script>** tag is used to tell the browser that a script follows

2.1 Internal

- The **<SCRIPT>** tag is used to **embed** JavaScript code in HTML documents

<SCRIPT LANGUAGE="JavaScript">

[JavaScript Statements...]

</SCRIPT>

- JavaScript can be placed **anywhere** between **<HTML>** and **</HTML>** tags
- two possibilities are the **<HEAD>...</HEAD>** portion and the **<BODY>...</BODY>** portion

2.1 Internal

<HTML>

**<HEAD><TITLE>Using Multiple
scripts</TITLE>**

<SCRIPT LANGUAGE="JavaScript">
[JavaScript statements...]
</SCRIPT>

<SCRIPT LANGUAGE="JavaScript">
[JavaScript statements...]
</SCRIPT>

</HEAD>

<BODY>

<H1>This is another script</H1>

<SCRIPT LANGUAGE="JavaScript">
[JavaScript statements...]
</SCRIPT>

</BODY></HTML>

2.2 External

- We place script in a **separate file** and include this in HTML code
- **SRC** attribute of the **<SCRIPT>** is used to include the external JavaScript file in HTML

<script src="myscripts.js"> </script>

- Are useful when you have **lengthy scripts**
- Improve the **readability**

3. JavaScript

- **Using the Semicolon**

```
document.write("Hello"); alert("Good  
bye") document.write("Hello")  
alert("Good bye")  
document.write("Hello"  
); alert("Good bye");
```

- **Case Sensitivity**

- **Comments:**

- single line //
- Multiple lines /* */

3. JavaScript

- **Using Quotes**

- `document.write("Hello
World")`

- `document.write("Hello
World")`

4. Writing

```
<HTML>
<HEAD>
<TITLE>Hello world.</TITLE>
</HEAD>
<BODY>
<script language="javascript">
document.write("<h1>Hello world</h1>");
</script>
</BODY>
</HTML>
```

Start of JS script

Writing on webpage

End of JS script

HTML code in JS

4. Writing



4.1 Variables in

- Variable is the **name** of a **memory location** which **holds** the data of a **certain type (data types)**
- There are **four common** data types in JavaScript
numbers, strings, Boolean, null values
- JavaScript is a **loosely typed** language

4.1 Variables in

- The word **“var”** is used to declare a variable
 - `var LastName = “Smith”`
 - `var AccountNumber = 1111`
- Variable Naming
 - First character can not be a **digit**
 - Other characters may be **digits, letters** or **underscore**
 - **Case sensitive**

4.1 Variables in

- **Variable Initialization**
 - **var** *variableName* = *initialValue*
 - **var** variableName1 =
initialValue1, variableName2 =
initialValue2, ...

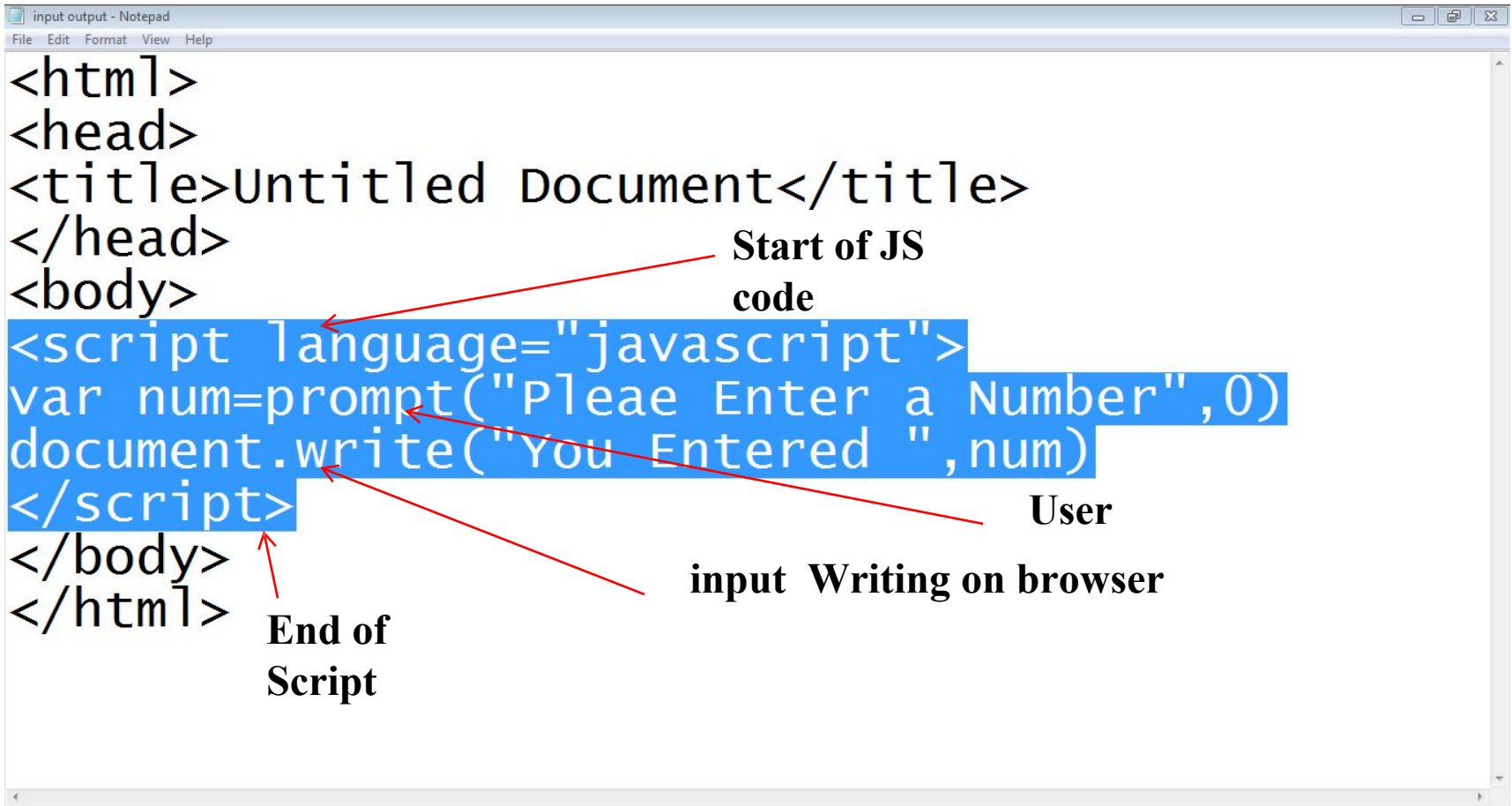
5. JavaScript

- An operator is simply a **symbol** that tells the compiler (or interpreter) to perform a **certain** action
- **Assignment Operator:** =
- **Arithmetic Operators:** +, -, *, /, %, ++, --
- **Logical Operators:** &&, ||, !
- **Comparison Operators:** ==, ===, !=, !==, <, >, <=, >=

6. Input Out put in

- **write();** is used to write on browser
 - **document.write(“hello world”)**
 - **document.write(a)**
- **prompt();** is used to take input from users
 - **var num = prompt(“Please Enter a Number”, 0)**

6. Input Out put in



The image shows a Notepad window titled "input output - Notepad" containing the following code:

```
<html>
<head>
<title>Untitled Document</title>
</head>
<body>
<script language="javascript">
var num=prompt("Pleae Enter a Number",0)
document.write("You Entered ",num)
</script>
</body>
</html>
```

Annotations with red arrows point to specific parts of the code:

- Start of JS code** points to the opening `<script language="javascript">` tag.
- User** points to the `prompt` function in the JavaScript code.
- input Writing on browser** points to the `document.write` function in the JavaScript code.
- End of Script** points to the closing `</script>` tag.

7. JavaScript

- **User defined functions**
- **Predefined functions**

7. JavaScript

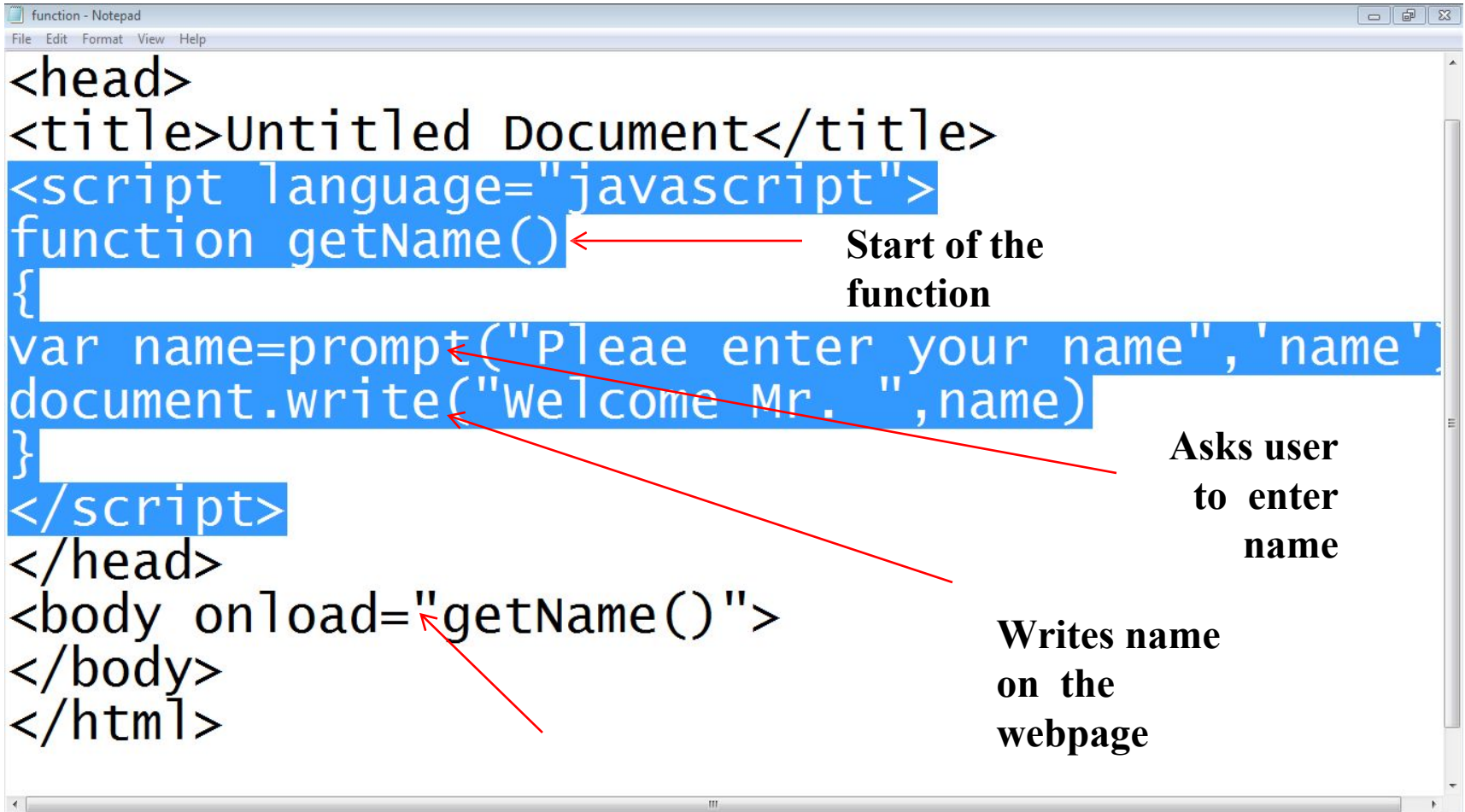
- Functions are defined using the keyword **function**, followed by the **name** of the function and **list of parameters**

```
function functionName([parameters])  
{  
  [statements]  
}
```

7. JavaScript

- **Calling a function**
 - The syntax of a function call is:
functionName([arguments])

7. JavaScript



The image shows a Notepad window titled 'function - Notepad' containing the following HTML and JavaScript code:

```
<head>
<title>Untitled Document</title>
<script language="javascript">
function getName()
{
var name=prompt("Pleae enter your name",'name');
document.write("Welcome Mr. ",name);
}
</script>
</head>
<body onload="getName()">
</body>
</html>
```

Annotations with red arrows point to specific parts of the code:

- An arrow points to the opening curly brace of the `function getName()` block, with the text **Start of the function**.
- An arrow points to the `prompt` function call, with the text **Asks user to enter name**.
- An arrow points to the `getName()` call in the `onload` attribute, with the text **Writes name on the webpage**.

Calling a function

7. JavaScript

- **Common events**
- **onClick**
- **onDblClick**
- **onChange**
- **onFocus**
- **onMouseOver**
- **onMouseOut**
- **onSubmit**
- **onload**

7. JavaScript

- **Some common predefined math functions**
 - **Math.sqrt**
 - **Math.pow**
 - **Math.abs**
 - **Math.max**
 - **Math.min**
 - **Math.floor**
 - **Math.ceil**
 - **Math.round**
 - **Math.random**

8. Conditional

- **If statement**
 - **if**
(condition)
statement
 - **if(condition)**
{ statements }
- **If-else statement**
 - **if(condition)**
{statement}
else
{statements}

8. Conditional

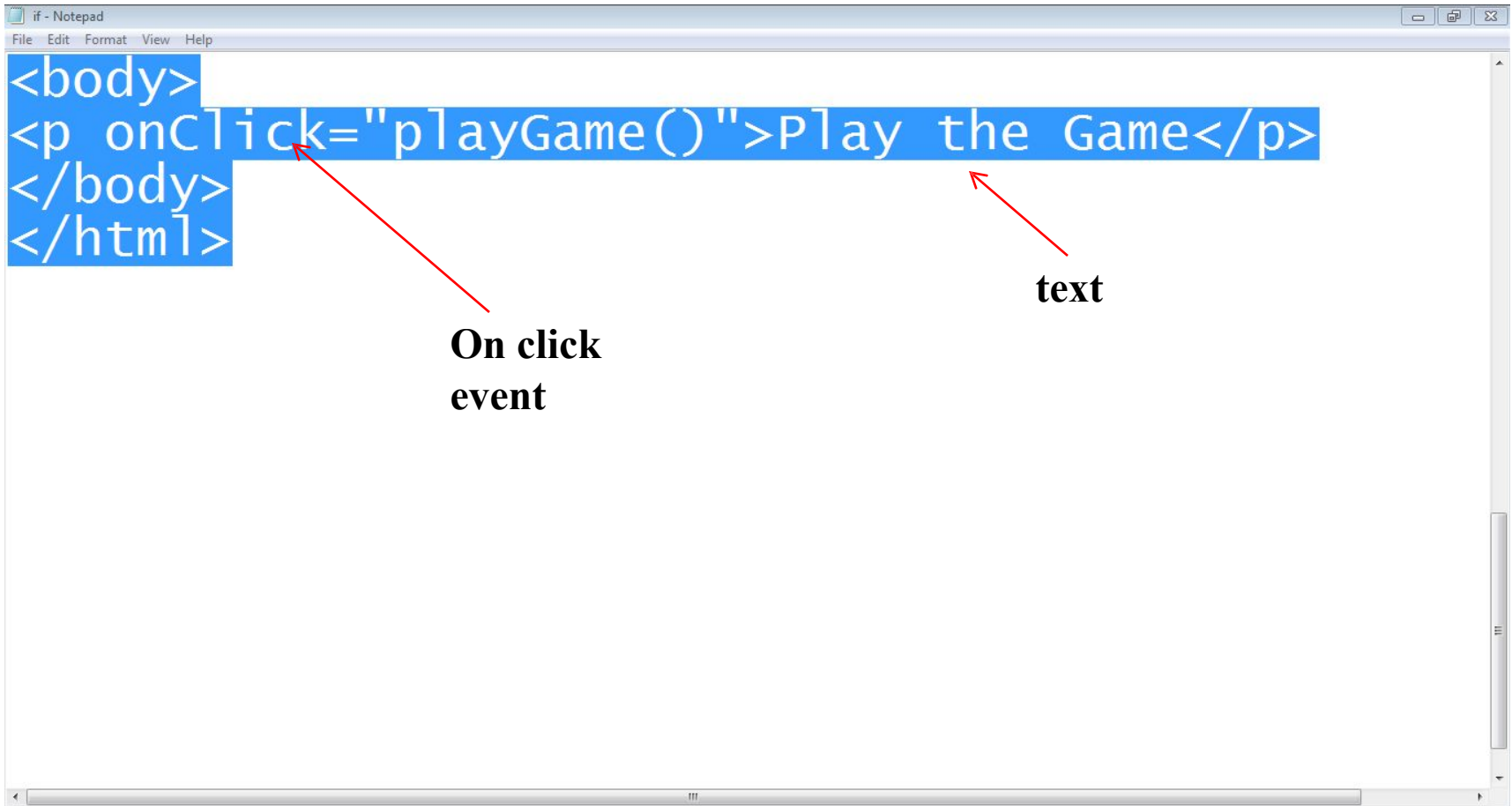
```
if - Notepad
File Edit Format View Help
<title>Using If condition</title>
<script language="javascript">
function playGame()
{
var res=parseInt(Math.random()*10)
var num=prompt("Pleae a number",0)
if(num==res)
document.write("You Won")
else
document.write("Your loss, correct Answer is"
,res)
}
</script>
```

Random number is generated

User's Input

If condition

8. Conditional



The screenshot shows a Notepad window titled 'if - Notepad' with a menu bar (File, Edit, Format, View, Help). The text area contains the following HTML code:

```
<body>  
<p onClick="playGame()">Play the Game</p>  
</body>  
</html>
```

Two red arrows point from text labels to the code:

- An arrow points from the label **On click event** to the `onClick="playGame()"` attribute in the `<p>` tag.
- An arrow points from the label **text** to the text `Play the Game` inside the `<p>` tag.

9.

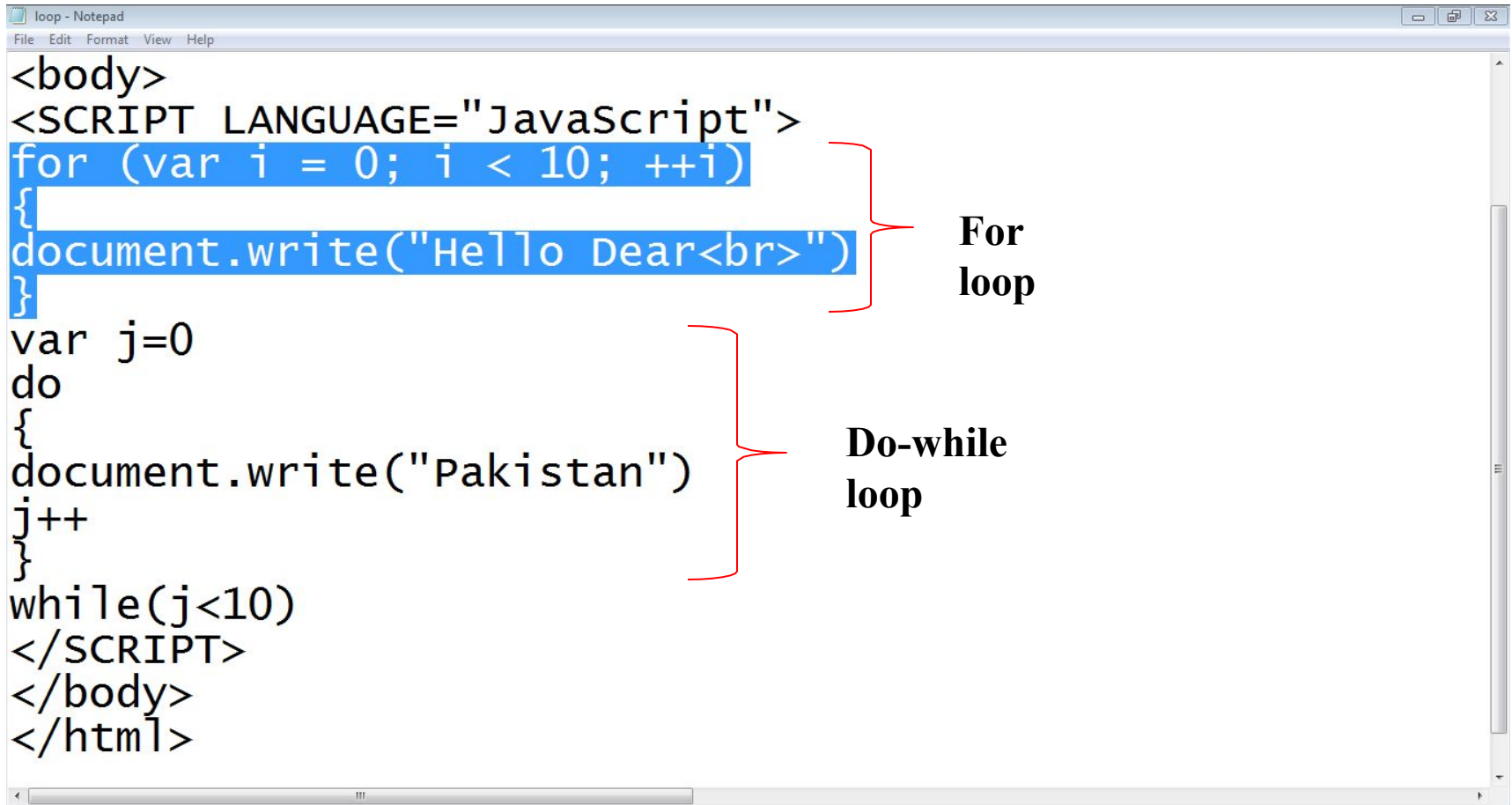
- **For loop**

```
for(var i=1; i<10; i++)  
{  
  Document.write("hello  
world")  
}
```

- **While loop**

```
While(condition)  
{  
}
```

9.

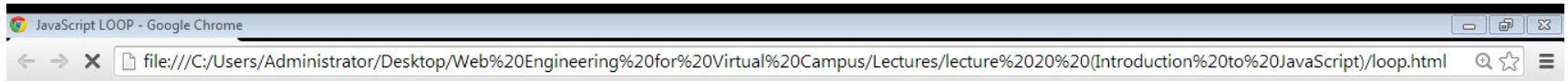


```
loop - Notepad
File Edit Format View Help
<body>
<SCRIPT LANGUAGE="JavaScript">
for (var i = 0; i < 10; ++i)
{
document.write("Hello Dear<br>")
}
var j=0
do
{
document.write("Pakistan")
j++
}
while(j<10)
</SCRIPT>
</body>
</html>
```

For loop

Do-while loop

9.



Hello Dear!
Hello Dear!
Hello Dear!
Hello Dear!
Hello Dear!
Hello Dear!
Hello Dear!
Hello Dear!
Hello Dear!
Hello Dear!

**Output of for
loop**

PakistanPakistanPakistanPakistanPakistanPakistanPakistanPakistanPakistanPakistan

**Output of
do- while
loop**

Summar

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Reference

- **Chapter 11.** Beginning HTML, XHTML, CSS, and JavaScript, by Jon Duckett, Wiley Publishing; 2009, ISBN: 978-0-470-54070-1.
- **Chapter 2,** Learn JavaScript, by Chuck Easttom, Wordware Publishing; 2002, ISBN 1-55622-856-2